Glossary of Explore NPX file Output

Term	Description	Utility in analysis	Additional comments
SampleID	Sample name as obtained from customer manifest	Sample identification	
Index	Number corresponding to the index sequence used to label each sample in Olink workflow	Minimal, as the sample is already identified by the sampleID at the NPX file level	
OlinkID	Assay (protein being evaluated in a particular line); unique identifier on the vendor side	Minimal, mostly for software recognition of each each	
UniProt	Six-digit alphanumeric code derived from UniProt data base. Assays that lack a UniProt ID well have distinct syntax	This enables rapid lookup of significant results obtained in anaylsis. Naturally, it's difficult to have a strong command of 1400+ protein, Uniprot can help identify some biology Olink may have discovered	Refer to https://www.uniprot.org/

Assay	Common protein name of the protein being assayed by Olink	In addition to sampleID, this is the main descriptor of the dataset	
MissingFreq	For a given assay, the fraction (out of 1) of data beneath the limit of detection (see below)	Helps in identify low expression proteins.	Multiply by 100 for percent of data missing. 0 or 0% means all data was above LOD
Panel	Which panel a given assay is on	Useful to know if a large portion of proteins of interest are in a specific panel	Either Cardiometabolic, Inflammation, Neurology, or Oncology as of early 2021.
PlateID	Identifier of which plate the sample and assay was run on		Generated by the Novaseq
QC Warning	Binary; identifies samples that have passed Olink QC	Olink recommends excluding the data with QC warning in your data analysis	For details around Quality Control, review section 2 in the Certification of Analysis
LOD	Limit of detection; result of three standard deviations beneath the mean NPX value for a	Olink recommends that assays with a large proportion of samples below LOD is excluded from the analysis. The limit for exclusion should be decided on a study basis and consider design including sample size and experimental variables. Suitable exclusion limits may be in the range of	

	given assay; in Olink Explore, LOD is calculated for each plate individually	less than 25-50% of samples above LOD. For mor information refer to: https://www.olink.com/question/how-is-the-limit-of-detection-lod-estimated-and-handled/	
NPX	Normalized protein expression; calculated for each assay based on internal and external controls	Log 2 scale; statistically ready for simpler approaches like T-test/ANOVA as well as higher dimensional approaches like prinicipal component analysis, neighborhood embedding, etc	For further information around NPX please refer to: https://www.olink.com/question/what-is-npx/